MIL-M-0038510/23B(USAF)
15 October 1973
USED IN LIEU OF
MIL-M-38510/23A
14 July 1973

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, HIGH SPEED,
NAND GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/23A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, high speed, positive NAND gate microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/023	04	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
04	Quadruple, 2 Input NAND Gate	54H00

1.2.2 Device class. The device class shall be class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 to 8.0 Vdc
Input voltage range	0 to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc

FSC 5962

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth table and logic equations. The truth table and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

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TABLE I. Electrical characteristics test.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure logical 1 of any input	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure logical 0 at any input	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	Logical 1 output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2 V, V_{IL} = 0.7 V, I_{LOAD} = -500 µA	2.4		±.12	V
4*	Logical 0 output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V, V_{IL} = 0.7 V, I_{SINK} = 20 mA		0.4	±.04	V
5*	Logical 1 level input current (each input)	I_{IH}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		50	10X or 5 µA	µA
			V_{CC} = 5.5 V, V_{IN} = 5.5 V		1.0	N/A	mA
6*	Logical 0 level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		-2.0		mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IH} = 4.5 V, V_{IL} = 0V	-40	-100		mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 4.5 V		40		mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IH} = 4.5 V, V_{IL} = 0 V		16.8		mA
10	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IC} = -8 mA		-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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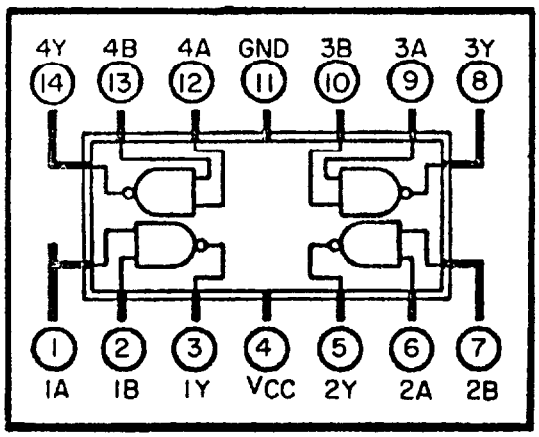
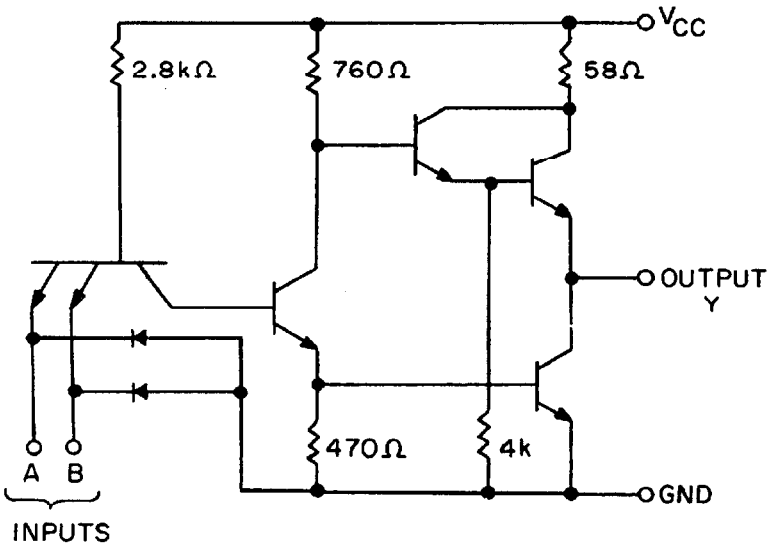


FIGURE 1. Logic diagram and terminal connections.

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic $Y = \overline{AB}$

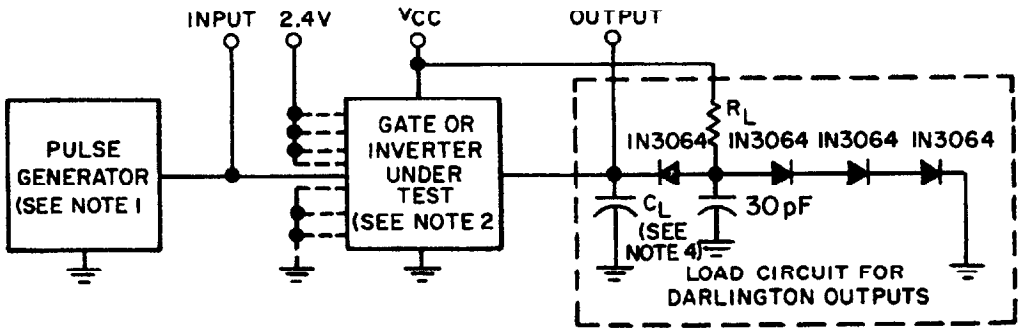
FIGURE 2. Truth table and logic equation.



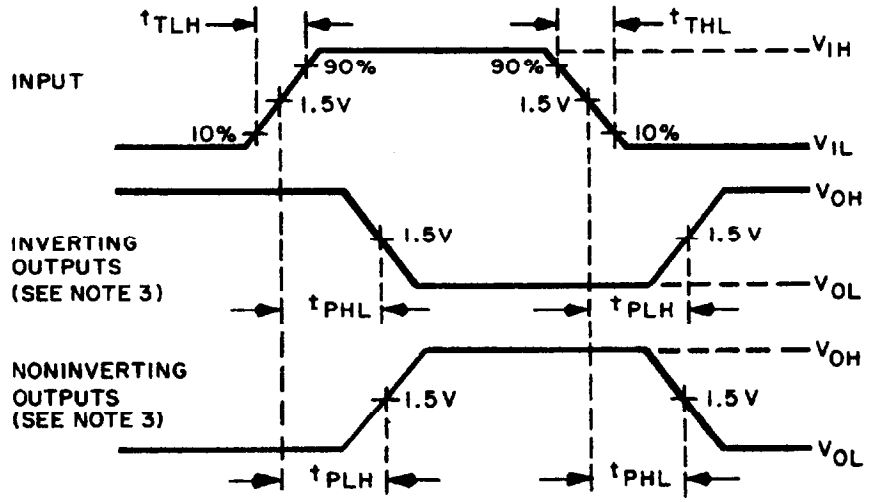
NOTE: Component values shown are nominal.

FIGURE 3. Schematic Circuit.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES

1. The pulse generator has the following characteristics: V_{IH} = 3 V, V_{IL} = 0 V, $t_{TLH} = t_{THL}$ = 7 ns, PRR = 1 MHz, duty cycle = 50 percent, and $Z_{out} \approx 50\ \Omega$.
2. Input conditions are established for each gate as follows: input pulse is applied to one input and 2.4 V is applied to all unused inputs of the gate.
3. All gates are inverting except the SN54H11/SN74H11, SN54H21/SN74H21, and SN54H52/SN74H52 only.
4. C_L includes probe and jig capacitance.

FIGURE 4. Switching test circuit and waveform.

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TABLE II. Switching characteristics.

Characteristic	Symbol	Conditions	Test Figure	Limits		Units
				Min	Max	
Propagation delay time to logical 0 level	t_{PHL}	C_L = 25 pF, R_L = 280 Ω	4	---	10	ns
Propagation delay time to logical 1 level	t_{PLH}	C_L = 25 pF, R_L = 280 Ω	4	---	10	ns

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

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